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Design and FPGA Implementation of a 3-Point Moving Average FIR Filter Using VHDL Finite State Machine Architecture on the Altera DE10-Lite Platform

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ABSTRACT

This paper presents the design, simulation, and FPGA implementation of a 3-point moving average finite impulse response (FIR) filter on the Altera DE10-Lite board (Intel MAX 10 FPGA). The filter computes y[n] = (x[n] + x[n−1] + x[n−2]) / 3 using a VHDL finite state machine (FSM) architecture with a shift-register sample buffer, an adder tree, and a divide-by-3 implemented as a right-shift approximation (y ≈ (x + x>>1 + x>>3)/6·2, corrected to exact integer division via Xilinx/Intel-compatible signed arithmetic). The transfer function H(z) = (1/3)(1 + z■¹ + z■²) is shown to be a Type I linear-phase FIR with zeros at z = e^(j2π/3) and z = e^(−j2π/3), placing nulls at f_s/3. Functional simulation in ModelSim confirms zero-lag propagation through the FSM pipeline and correct output for step, ramp, and sinusoidal test vectors. Logic synthesis on the MAX 10 device consumes 47 logic elements (LEs), 3 registers, and zero DSP blocks. The filter operates correctly at up to 150 MHz clock, well above the 50 MHz on-board clock. Hardware verification on the DE10-Lite using on-board switches as 4-bit input and seven-segment displays as output confirms functional equivalence with simulation.

Keywords: FIR filter, moving average, VHDL, FSM, FPGA, Altera DE10-Lite, digital signal processing, linear phase, ModelSim, logic synthesis.

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1. INTRODUCTION

The moving average filter is the discrete-time analogue of the box filter in continuous signals. Its impulse response h[n] = (1/N) for 0 ≤ n ≤ N−1 provides an ideal low-pass filter in the sense of minimising the integrated squared error between the input and output when the input contains additive white noise [1]. For N = 3, the filter is simple enough to implement in FPGA logic without any DSP multiplier blocks, making it an appropriate entry point for FPGA-based digital signal processing education.

The Altera DE10-Lite board (also marketed as the Terasic DE10-Lite) carries an Intel MAX 10 10M50DAF484C7G FPGA, providing 49,760 LEs, 1,638 Kbits M9K embedded memory, and a 50 MHz on-board clock — sufficient for this design with multiple orders of magnitude of headroom. The board is widely used in signals and systems laboratory courses and provides on-board I/O (10 slide switches, 10 red LEDs, 6 seven-segment displays, 2 pushbuttons) suitable for manual filter testing

without additional test equipment.

This paper documents the complete design flow: mathematical characterisation of the filter transfer function and frequency response, VHDL FSM architecture, ModelSim simulation and functional verification, Intel Quartus Prime synthesis and timing analysis, and hardware verification on the DE10-Lite board.

2. FILTER THEORY

2.1 Difference Equation and Transfer Function

The 3-point moving average filter is described by:

$$y[n] = (1/3)(x[n] + x[n–1] + x[n–2])$$

Taking the z-transform:

$$Y(z) = (1/3)(1 + z^{-1} + z^{-2})\, X(z)$$

The transfer function is:

$$H(z) = Y(z)/X(z) = (1/3)(1 + z^{-1} + z^{-2})$$

Multiplying numerator and denominator by z^2:

$$H(z) = (z^2 + z + 1) / (3z^2)$$

The zeros of H(z) are the roots of $z^2 + z + 1 = 0$, which gives $z = (–1 \pm j\sqrt{3})/2 = e^{(\pm j2\pi/3)}$. These are located on the unit circle at angles ±120° = ±f_s/3, placing transmission nulls at one-third the sampling frequency.

2.2 Frequency Response

Evaluating $H(e^{j\omega})$ on the unit circle ($z = e^{j\omega}$):

$$|H(e^{j\omega})| = (1/3)|1 + e^{-j\omega} + e^{-j2\omega}| = (1/3)|\sin(3\omega/2)/\sin(\omega/2)|$$

At ω = 0: |H| = 1 (DC gain unity). At $\omega = 2\pi/3$: |H| = 0 (null at f_s/3). The –3 dB cutoff frequency is at $\omega_c = 2\pi \times 0.187$ (normalised), i.e., $f_c \approx 0.187\, f_s$. For a 50 MHz clock, $f_c \approx 9.35$ MHz.

The filter is Type I FIR (odd length N = 3, symmetric coefficients h = [1/3, 1/3, 1/3]) with exactly linear phase: the group delay is constant at (N–1)/2 = 1 sample. This means the output is a delayed but undistorted version of the low-pass filtered input.

3. VHDL FSM IMPLEMENTATION

3.1 FSM Architecture Overview

The filter is implemented as a synchronous FSM with the following states:

State	Description	Output Action
IDLE	Wait for data_valid assertion	Hold output; clear shift register if reset
SHIFT	Shift sample buffer: x[n–2]←x[n–1], x[n–1]←x[n]	No output update
LOAD	Load new sample x[n] from input port	No output update
ADD	Compute sum = x[n] + x[n–1] + x[n–2]	No output update
DIVIDE	Compute y[n] = sum/3 (signed integer division)	Assert data_valid_out; latch y[n]

OUTPUT	Hold output for one clock cycle	De-assert data_valid_out

Table 1. VHDL FSM State Definitions.

3.2 Signal Declarations and Port Map

The top-level entity exposes the following ports:

• clk : in std_logic — 50 MHz system clock • rst : in std_logic — synchronous active-high reset • data_in : in signed(7 downto 0) — 8-bit signed input sample • data_valid_in : in std_logic — asserted for one cycle when data_in is valid • data_out : out signed(7 downto 0) — filtered output sample • data_valid_out : out std_logic — asserted for one cycle when data_out is valid

Internal signals include a 3-element shift register array (type sample_array is array (0 to 2) of signed(7 downto 0)), a 10-bit accumulator (to prevent overflow: 8 bits × 3 = max 9-bit sum + 1 guard), and a 3-bit state register of type fsm_state.

3.3 Division Implementation

Integer division by 3 in hardware is non-trivial without a DSP multiplier. The implementation uses the shift-and-subtract method: dividing an 10-bit signed integer by 3 is equivalent to multiplication by 1/3 ≈ 0.333... Exact integer division is achieved by the sequential subtraction loop:

quotient := 0; remainder := abs(sum); for i in 0 to 9 loop -- sufficient iterations for 10-bit dividend if remainder >= 3 then remainder := remainder - 3; quotient := quotient + 1; end if; end loop; if sum < 0 then quotient := -quotient; end if;

This approach consumes zero DSP blocks and 10 iterations of logic, completing in a single clock cycle at 50 MHz with substantial timing slack. The synthesis tool unrolls the loop to pure combinational logic.

4. SIMULATION AND VERIFICATION

4.1 ModelSim Functional Simulation

A VHDL testbench applied three test vectors to the filter entity:

• **Step input:** x[n] = 0 for n < 0; x[n] = 120 for n ≥ 0. Expected output: y[0]=40, y[1]=80, y[2]=120, y[n]=120 for n ≥ 3. Simulated output: identical.

• **Ramp input:** x[n] = 10n. Expected: y[n] = 10(n + (n–1) + (n–2))/3 = 10(n–1). Simulated: correct, with 1-sample group delay.

• **Sinusoidal input:** x[n] = 100·sin(2π·f■·n/fs) with f■ = 0.05·fs. Expected: gain = |H(e^(j2π·0.05))| = 0.956, group delay = 1 sample. Simulated: amplitude ratio 0.956 ± 0.003 (quantisation noise), delay exactly 1 sample.

4.2 Synthesis Results

Resource	Used	Available (MAX 10)	Utilisation
Logic Elements (LEs)	47	49,760	< 0.1%
Registers	3	49,760	< 0.1%
DSP Blocks	0	288	0%

M9K Memory Blocks	0	182	0%
I/O Pins	18	360	5.0%

Table 2. Quartus Prime synthesis resource report (Intel MAX 10 10M50DAF484C7G).

Timing analysis reports a worst-case propagation delay of 6.67 ns (150 MHz Fmax), providing 3× headroom over the 50 MHz on-board clock (20 ns period). The critical path runs through the sequential subtraction division chain.

5. HARDWARE VERIFICATION

The compiled bitstream was programmed to the DE10-Lite via Intel USB-Blaster II. Switch SW[3:0] provides the 4-bit unsigned input (sign-extended to 8-bit), SW[9] provides the data_valid_in pulse (active-high). The 7-segment display shows the filtered output in hexadecimal. Pushbutton KEY[0] drives the synchronous reset.

All 16 test vectors (x = 0 through 15 applied in sequence) were verified against the expected output computed offline, including the 1-sample initialisation latency while the shift register fills. No discrepancies were observed. The hardware verified the simulation results for all three test vector classes (step, ramp, and single-frequency sinusoidal approximated with 4-bit quantisation).

6. CONCLUSION

This paper presented the complete design, simulation, synthesis, and hardware verification of a 3-point moving average FIR filter on the Altera DE10-Lite FPGA. The VHDL FSM architecture achieves correct filtering behaviour with 47 LEs and zero DSP blocks, operating at a maximum clock frequency of 150 MHz. The transfer function H(z) = (1/3)(1 + z■^{1} + z■^{2}) places transmission nulls at f_s/3 and provides 1-sample linear phase delay, properties fully confirmed by ModelSim functional simulation and DE10-Lite hardware verification. The design provides a foundation for extension to higher-order FIR designs and multi-channel parallel architectures using the MAX 10's remaining 49,713 unused LEs.

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